Hoapa

Logic, collective inquiry and open-science workshops

Jean-Charles Belouard

Helsinki Logic Group, 11 November 2026

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Jean-Charles Belouard: great engineer, ok driver

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Introduction — The context

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The core of my research consists in drawing inspiration from the philosophy of mathematics and the engineering of networks to formulate a theory of work constrained by “permacultural” logic: how to produce reusable results while preserving the conditions for future work. Applied to collectives, it becomes an extension of the scientific method which incorporates participatory democracy and sustainability.

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  $\rightarrow$ well-regulated network of relations among scientists, each retaining a well-regulated relation with Nature

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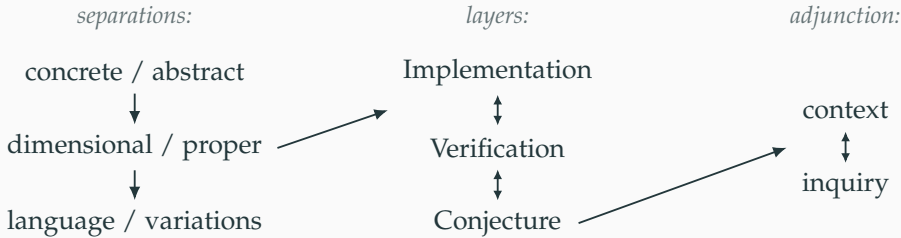
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`[ perspectivism, decentralization, autonomy, openness to peers, expressiveness, transparency, scalability, self-reflection... ]`

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…Hoapa as an invariant in the effective theories we see

Introduction — The main thesis

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My main thesis states that in a system that puts the dialectic between concrete and abstract to work (brain, computer, community), the relation between concrete substrate and logical labor must be a resource entirely dedicated to the design and application of representations and algorithms. A form of separation between two kinds of decomposition—analytic (as in prime factorization) and synthetic (as in vector coordinates relative to a basis)—is a guiding principle, informed by Lautman, which can keep this relation fruitful: in a community, between thinkers and the knowledge they produce.

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  engine vs. organism, the role of closure;
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The physical starting point is that work depends on constraints that channel energy into particular effects. In living systems, activity also helps maintain the machinery that makes further activity possible. Abstract representations and algorithms enable a system to retain and share experience, discover relations, construct and test possibilities, revise its own means of understanding and action, and turn results into conditions for further work. This makes the relation between concrete conditions and abstract operations a powerful resource. Lautman supplies a way to investigate such relations through the mathematical and physical constructions that give them substance; Peirce, Simondon, and Kauffman help connect the question to living beings, their environments, and their practices of understanding. Hoapa's proposed separation between thinkers and the organization of knowledge is an engineering approach to developing this potential.

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Constraints: effects — *Maintain*: closure — *Abstract*: secondary loop

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Hoapa is an open project in mathematics, software, and the design of collective inquiry. I set its direction, develop its questions, and decide which proposals to pursue. AI agents help explore mathematical connections, compare approaches, and write and review code and documentation. Cubical Agda checks that the formalized constructions fit together under explicit definitions and assumptions, allowing new work to build on results already established.

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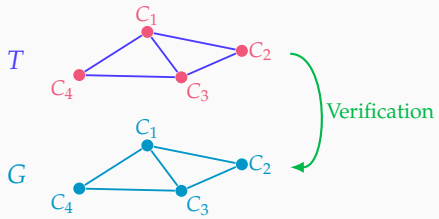
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Hoapa is designed as a network of small conversations in which one group's work becomes another group's starting point. A question brings participants together; their discussion produces a document that others can examine, challenge, or develop. These documents form a growing map of questions, concepts, arguments, and dependencies: a knowledge graph. So that contributions can differ and come unforced, the system gives participants control and equal standing (they all play by the same rules they themselves manage). Knowledge acquires structure independently of the structure of its producers, be it social, political, economic…

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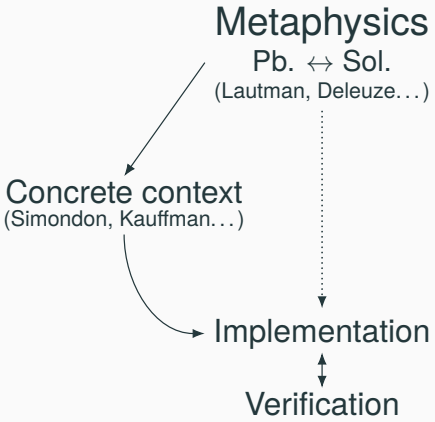
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- *Independently*: respecting the concrete/abstract separation of concerns and dedicated relation between the two

Introduction — How Hoapa works



Introduction — What is checked, what is conjectured

Verification

The communication protocols specify how these conversations proceed and how their results enter the shared documentation. Software is used as a referee and is itself subject to modification by the conversations. Alongside the conversation process, a mathematical verification layer is being developed in cubical Agda, a language in which proofs can be checked by a computer. One existing result establishes that consistently renaming participants cannot change the checker’s verdict. The broader aim is to make the system’s promises precise enough to test and prove.

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- *Verification*: machine-checks that the record of each conversation is well-formed according to the protocol, never that its conclusions are true
- *Participants*: the protocol is blind to identity of contributors, which means it can't count on specialization (is knowledge lost?)

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Conjecture

The project seeks a mathematical correspondence—an adjunction—between two poles. One organizes inquiry through conversations, verification, and knowledge graphs. The other represents mathematical contexts and the different perspectives through which they can be investigated. The conjecture is guided by an intuition prompted by the perspectival nature of the world: our different understandings of it invite discussion, without presuppositions, from which something the world can reuse may emerge. The conjecture asks whether a universal property can distinguish Hoapa's organization from alternatives. This talk will explore a candidate in which methods are represented by logics and mathematical worlds by the inner models they generate.

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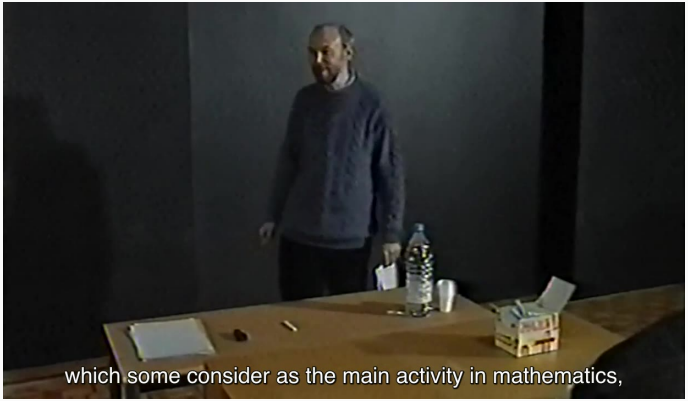
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- *Universal property*: "specialness" vs. optimality

Rising sea



P. Deligne, *L'œuvre d'Alexandre Grothendieck*, Nice, 1996

Permacultural logic

The project also looks for this pattern inside logic, where “permaculture” is understood broadly as the two-step process of (1) getting to know one’s context as much as possible, (2) acting minimally in a manner that will enhance one’s condition while preserving and enriching the context. A proof may begin with a provisional assumption (borrow), derive a conclusion (work), and turn the derivation into a conditional statement that later proofs can use (replay). What began as work under particular conditions becomes a reusable achievement. Hoapa investigates this borrow–work–replay pattern as a possible logical expression of its permacultural ambition: the harmony between receiving and giving back.

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- *Harmony*: as in Gentzen, type theory, Hilbert's axiomatic method
- *Replay*: the hidden part of logic that makes logic work

Development

The architecture of a “mature” Hoapa system is both demanding—to respect its constraints—and—by virtue of presuppositionlessness—unpredictable. Autonomous open-science workshops could make these ideas tangible and seed larger systems, embracing both pragmaticism and experimentalism. Their first requirement is usefulness: knowledge sharing must be truly at stake, so that the organization develops through serving a real purpose. Concretely, I envision something based on the philosophy of “document-driven-development”, where documentation gradually replaces the instructor, both as the joint source of truth and the manager of the process.

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- *Unpredictable*: collective inquiry helps determine how collective inquiry is organized
- *Stake*: what distinguishes and makes citizen assemblies efficient; you can't simulate life or understanding
- *Documentation*: does in the Lautmanian-analytic what specialization does in the synthetic agents: shared source and management, but "softer"

Implementation — Use cases

A park

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Here it is crucial to understand that the system can preserve its advantageous mathematical, social and engineering properties if it is **only** a communication system, and not a full management system which also enacts in reality what has been discussed in the abstract.

A park

Its task is to chart the world and design appropriate algorithms, not interfere with and certainly not control it. (That would kill all the carefully crafted separations of concerns, starting with the role-symmetry of agents, because it necessarily introduces some hierarchical structure and tension—for basically the same reasons you can share an idea with the entire world without losing it, but you can't do that with a loaf of bread.)

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- If it needs something it simply asks for it, and expects those needs to be met (for instance, if it uses the internet, it states via a self-reflective conversation how much bandwidth it needs, and specifies the characteristics of its network). Individuals or communities, outside of the protocol, would then respond to those needs without compensation.

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- If it needs something it simply asks for it, and expects those needs to be met (for instance, if it uses the internet, it states via a self-reflective conversation how much bandwidth it needs, and specifies the characteristics of its network). Individuals or communities, outside of the protocol, would then respond to those needs without compensation.
- If it wants something to be known it simply makes it as available and accessible as possible, which includes a pedagogical effort in presentation and terminology. But it can't advertise, at the risk of favoring certain targets over others.

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In either direction, we can think of Hoapa as a coach in sports: its domain of action is preparation and strategy, but it can never enter the field of play or control the players. The game belongs to the players.

A park

In this example, we can imagine an institution/Hoapa hybrid in which the city government has delegated the supervision of the park to the protocol, but it's still the paid workforce of gardeners which will perform the tasks suggested by the protocol. But ideally, in the spirit of the protocol, it would be only voluntary work, suggested by the protocol but decided at the level of the individuals and communities.

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Which makes Hoapa protocols especially well suited to organize the open source economy.

Open source

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The main recurring problem open source projects face is scalability and continuation beyond the circle of core contributors and maintainers, usually the instigators of the project: because open source projects are structured around “circles of knowledge and influence”, which is in essence a hierarchy, it is very likely to terminate the moment the inner circle quits.

Implementation — Use cases

Open source

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The same arguments can be used to suggest a reform of how science is conducted.

Implementation — Use cases

Science

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While the scientific method, in the domains where it is applicable, makes claims answerable to evidence and correction, it doesn't prevent other types of evils. Notably a mismanagement of efforts toward certain technologies at the expense of understanding, the fragmentation of the study of Nature into opposing departments, the proliferation of opaque jargons, the insistence that there exists a "view from nowhere", the enclosure of knowledge behind paywalls, proprietary tools, and restricted access, the depreciation of errors and failures, etc.

Implementation — Use cases

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- *Language*: "Learn once, write everywhere" ← a shared conceptual landscape built through explicit and systematic comparisons (interpretation, equivalence, coherence…)

Implementation — Use cases

Science

Hoapa protocols are designed for education, as part of their mandate to treat everybody on an equal footing. What is removed with the status is not the knowledge, but the hunt for the status, and at the same time it resituates the obligation to make one's specialized knowledge accessible to others.

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Crucially, each individual practitioner gets to decide what questions are worth pursuing, opening a path to restoring science's sense of purpose from the ground up.

And the organization of inquiry is itself something we can try to understand mathematically.

Implementation — Use cases

Mathematics

All of the above applies in mathematics as well, but here the connection goes further: mathematics also supplies the language in which the organization of mathematical work can itself be investigated. Hoapa's formalization uses Cubical Agda, bringing together homotopy type theory, computer-checked proofs, and a line of thought connecting Alexander Grothendieck's ideas about homotopy types and higher groupoids with Vladimir Voevodsky's univalent foundations.*

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- *Further*: a method of articulating a context can generate a new context for further inquiry (in math and logic, in generalized evolution)

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> Grothendieck also questioned the ecological consequences of scientific activity; Hoapa approaches that concern by asking how understanding a world can help us inhabit it.

Generalized permaculture

I believe Hoapa protocols can stage and target the kind of recursive and self-reflective patterns just mentioned, in the ecosystem, by generalizing the concept of permaculture and applying it to a vast majority of human endeavors. Permaculture could in effect become the default mode of being of humanity, following this very simple algorithm:

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- Make a very detailed and executable map of your environment, including your imprint on it.
- Use it to figure out what is best for both you and your environment, and do it (which becomes the default if individual freedom replaces “investability” as the condition for action).

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> To tend a place is also to attend to the relations that sustain it—and those relations lead us into a network.

Implementation — Use cases

Networks

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A mature protocol regulating a park would necessarily have typed conversations with other park protocols, especially those close by, to share information and reuse each other's research. It would also have ties to all the independent but interlinked networks a city would be made up of: electricity grid, water management, transportation, agriculture… and then beyond the city limits, into all sorts of other communities organized through protocols, each bringing its own point of view, from playground design to natural philosophy.

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A park's investigation of soil and water becomes useful to another park; a software improvement made for its documentation becomes available to other projects; a scientific question raised locally finds research communities elsewhere equipped to pursue it. The park community's teaching material can help newcomers elsewhere. And communities can help one another translate documentation, test proposals, or investigate disagreements. Records of failed attempts can spare others the same mistakes.

Implementation — Use cases

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- *Care*: Hoapa's generalization of attachment theory into a two-step cyclic algorithm

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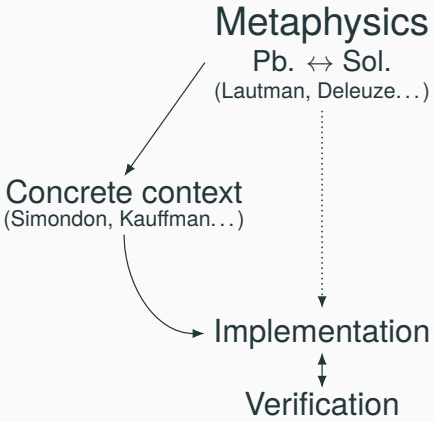
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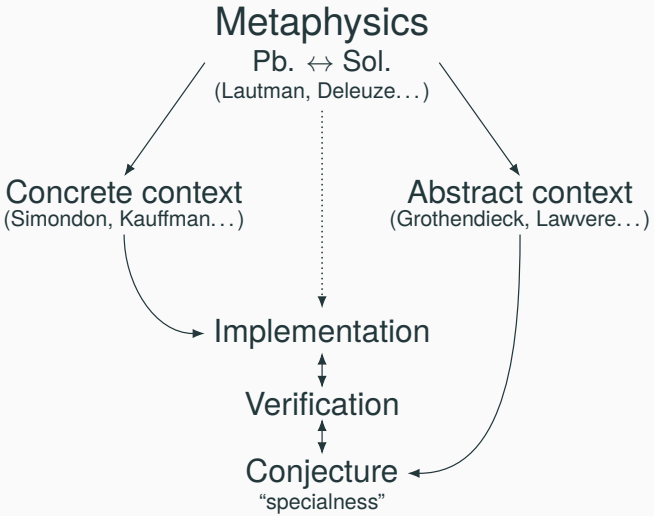
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- *Care*: Hoapa's generalization of attachment theory into a two-step cyclic algorithm
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  2. In turn, those newer elements become mature and take care of a new generation of elements

The Hoapa conjecture — an adjunction



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Motivations and goals

To motivate its adoption and distinguish Hoapa from other "interesting projects" we can attach buzzwords to, I need a mathematical reason for organizing inquiry this way. The goal is to characterize Hoapa by a universal property, in a setting that represents both the contexts it investigates and alternative ways of investigating them.

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If the conjecture succeeds, this universal property would translate into a claim of optimality for any implementation satisfying its conditions. The world we inhabit must explain why those conditions matter. Hoapa is also its own experiment: implementing its protocols lets us test the conjecture's claims.

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- *Optimality*: the constraints imposed on a process, in a given context, to achieve an intended outcome
  constraints = the protocol; process = collective inquiry; context = this world;
  outcome = understanding that enables generalized permaculture to be enacted

Motivations and goals

Perspectivism motivates this ambition; presuppositionlessness guides the response. The world has already produced different beings, experiences, skills, and ways of understanding. Their contributions must be able to enter the inquiry without their worth being decided in advance by a hierarchy or an established division of knowledge. What the world has already developed becomes material for collective work; what that work will produce remains open.

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- *Perspectivism*: global understanding assembled from local views, with obstructions to their integration retained as information
- *Presuppositionlessness*: common rules, open rules with equal power to revise them, and only the rules needed to make communication possible

Motivations and goals

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My larger hypothesis is that this movement participates in the evolution of the world itself. What Hoapa preserves as a representation and reflection of the context in which it is implemented, and the creativity it enables as a response to the problems posed by this context, form the basis for a new context in which a community of "synthetic" agents can work on an "analytic" representation of their world. And this statement can travel through the boundaries between concrete (theory of work) and abstract (the conjectured adjunction).

Categories from metaphysics

Lautman looked for the problems that recur beneath different mathematical theories: local and global, the whole and its parts, structure and existence. Zalamea reads his account of the passages between local and global, and of the intermediate structures that carry them, as anticipating sheaf theory. Category theory makes such recurrences explicit, through constructions, transformations and universal properties that can be compared from one theory to another. That is how this project passes from metaphysical problems to effective mathematics.

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The project has affinities with formalisms well adapted to the tasks it must carry out. Topos theory gathers local descriptions, the conditions under which they fit together, and an internal language in which to reason about them. Higher structures keep the different comparisons between descriptions, and let us ask how those comparisons fit together. Univalence, at the heart of homotopy type theory, treats equivalence as identity, so that properties and constructions transport between equivalent types.

The Hoapa conjecture — an adjunction

Categories from metaphysics

The proposed adjunction relates a category of contexts to a category of organizations of inquiry: write **Ctx** for the first and **Inq** for the second. We seek functors $F\colon \mathbf{Ctx} \to \mathbf{Inq}$ and $G\colon \mathbf{Inq} \to \mathbf{Ctx}$ with natural bijections $\mathrm{Hom}_{\mathbf{Inq}}(Fc, i) \cong \mathrm{Hom}_{\mathbf{Ctx}}(c, Gi)$. The unit $\eta_c\colon c \to GFc$ and the counit $\varepsilon_i\colon FGi \to i$ compare each side with its round trip. The mathematical form of **Ctx** is still under investigation, and neither functor has been constructed.

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- ε: Anthropic loss, the possible failure of the reconstruction map $\varepsilon_i\colon FGi \to i$ to be invertible; compare the gap between elementary equivalence and isomorphism, with Ehrenfeucht–Fraïssé games describing logical indistinguishability

Permacultural logic — From work to reusable construction

The permacultural question is how an achievement becomes a means of further construction. Implication provides a precise mathematical example. We borrow an assumption A, derive B, and *return* the derivation as an implication. Other proofs can *replay* that work when its conditions are supplied.

- *Return*: abstraction turns $w\colon \Gamma \times A \to B$ into $w^\sharp\colon \Gamma \to A \Rightarrow B$.
- *Replay*: evaluation applies the returned operation to an input: $(A \Rightarrow B) \times A \to B$.

Permacultural logic — From work to reusable construction

In the cartesian closed interpretation of intuitionistic logic, this is the adjunction:

$$(- \times A) \dashv (A \Rightarrow -) \qquad \mathrm{Hom}(\Gamma \times A, B) \cong \mathrm{Hom}(\Gamma, A \Rightarrow B)$$

The remaining assumptions Γ stay visible: discharging one dependency does not erase the others. Introduction and elimination are governed by *harmony*. Their compatibility makes work under an assumption interchangeable with a reusable operation expecting that assumption.

- *Harmony*: β and η make currying and uncurrying inverse—elimination extracts neither more nor less than introduction warrants.

Permacultural logic — What does the work leave available?

The same adjunction produces two familiar constructions. The *state monad* composes computations by passing the state left by one into the next. The *store comonad* holds a family of values together with a selected position, and allows that family to be considered from each position.

- *State monad*: $\mathrm{State}_A(X) = A \Rightarrow (X \times A)$—receive a state; return a result and a resulting state.
- *Store comonad*: $\mathrm{Store}_A(X) = (A \Rightarrow X) \times A$—retain an indexed family and a current position.

These structures organize work and access to its setting. Their laws alone do not preserve the conditions for further work. A computation $p(a) = (x, a_0)$ can replace every incoming state with the same a_0, erasing distinctions that subsequent work needs.

Permacultural logic — What does the work leave available?

Hoapa therefore asks about *continuation*: can the work we intend to do next still be performed from what has been returned? A publication may be sufficient for one use and insufficient for another. The requirement must specify what is to remain possible.

- *Continuation*: $q' \circ w = \bar{w} \circ q$—$q$ and q' publish before and after the work w; $\bar{w}$ performs the corresponding operation using only the published input.

Permacultural logic — The economy of replay

Earlier I called replay "the hidden part of logic that makes logic work." *Cut* gives this phrase a precise sense. An established result enters a later proof; cut elimination can remove that intermediate use without changing what is provable. But removing it can make the proof enormously longer.

- *Cut*: a proved result becomes a premise of further reasoning.

Permacultural logic — The economy of replay

There are families of short proofs with cut whose shortest cut-free proofs grow beyond every elementary bound in the original proof lengths. Reuse can therefore be dispensable for provability and decisive for *feasibility*.

- *Feasibility*: the same conclusions can require radically different proof lengths; preserving intermediate achievements can make complex work practicable.

This gives the sustainability intuition a concrete dimension. Mathematical development concerns what our means of reasoning let us accomplish with finite resources. An achievement can leave the set of consequences unchanged while transforming our capacity to establish and use them.

Permacultural logic — Achievements become means of inquiry

A further step occurs when an achievement changes the tools through which inquiry proceeds. In proof by *reflection*, a procedure is defined, its correctness is proved, and that proof supports its use in subsequent reasoning. Coq's `ring` tactic provides a concrete example: established mathematics becomes machinery for later mathematics, while the existing kernel continues to check the proofs.

- *Reflection*: a verified procedure becomes part of the proof-producing toolbox, without becoming a new trusted rule.

The reusable material can also widen: from derivations in Gentzen's systems, to formalized theories in Hilbert's metamathematics, to constructions and their transport between equivalent types in HoTT. Lautman's passage from structure to genesis, and Simondon's account of produced structures supporting further structuring, guide our interpretation of this movement.

Permacultural logic — Achievements become means of inquiry

Renewal occurs when achievements change what further work can do. With Kauffman, we must also ask whether the organization producing them maintains its *conditions of activity*. Hoapa's evolutionary hypothesis joins these demands: work should produce reusable achievements while sustaining and developing the conditions for further work.

- *Renewal*: preserve declared earlier uses and make a genuinely new operation available.
- *Conditions of activity*: reusable results still depend on people, skills, records, tools, and material support.